Acute Rheumatic Fever & Rheumatic Heart Disease

A Structured Clinical Reference — 27 Questions with Detailed Answers

Compiled June 2026 · Incorporates 2022–2026 evidence (GOAL/GOALIE trials, 2025 sex-difference meta-analysis, GAS vaccine pipeline) · Guideline anchors: 2020 RHDAustralia (3rd ed.), 2015 revised Jones criteria, 2020 ACC/AHA & 2021 ESC valve and 2018 ESC pregnancy guidelines, 2023 WHF.

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1. Risk factors for ARF/RHD — detailed

ARF is a delayed, immune-mediated sequela of group A streptococcal (GAS) pharyngitis in a genetically susceptible host. The dominant drivers are **social/environmental**, acting through repeated GAS exposure and inadequate treatment.

Domain	Specific risk factors
Socio-economic (dominant)	Poverty, household overcrowding, poor/inadequate housing, limited sanitation and hygiene, malnutrition, poor access to primary care (untreated sore throats), under-resourced health systems.
Host / demographic	Age 5–15 yr (peak of first ARF); female sex for progression to clinical RHD (see Q5); prior ARF (recurrence risk very high); family history.
Microbial / exposure	Recurrent or untreated GAS pharyngitis; "rheumatogenic" emm types; in highly endemic settings, GAS pyoderma/impetigo may also prime the immune response.
Environmental	Cooler / highland climate and seasonal crowding favour GAS transmission; institutional/boarding settings.
Genetic	HLA class II and several non-HLA immune genes (see below).

Why might the Northern zone (Manyara, Arusha, Singida) appear disproportionately affected?

There is no single definitive Tanzanian study proving a true northern-zone predominance, so this is best framed as a *plausible clustering of recognised risk factors* rather than an established epidemiological fact (referral patterns to the Jakaya Kikwete Cardiac Institute may also bias the apparent geographic distribution). Likely contributors in these regions:

- **Cooler highland climate** — Arusha and Manyara sit on the northern highlands and Singida on the central plateau; lower temperatures and indoor crowding favour respiratory GAS spread.
- **Pastoralist / semi-nomadic communities (e.g. Maasai, Barabaig)** — large households, communal living, and seasonal congregation increase transmission.
- **Lower access to prompt pharyngitis treatment and to secondary prophylaxis** — geographic dispersion and fewer

primary-care contacts mean more untreated GAS and more recurrences.

- **Socio-economic gradient** — overcrowding, poverty and undernutrition concentrate in these communities.

> Context: a Tanzanian school screening study reported sub-clinical RHD prevalence of ~2.1% (and ~34/1000 in a Dar es Salaam cohort), consistent with sub-Saharan African norms; RHD accounted for ~30% of cardiac surgery at JKCI in 2017–2018. The northern-zone observation deserves a dedicated geospatial study to confirm.

Is there a role for genetic predisposition? — Yes (permissive, not sufficient)

Genetic susceptibility shapes *who* develops ARF after GAS exposure (twin/family data suggest substantial heritability), but it requires the streptococcal trigger.

- **HLA class II** associations: HLA-DR7, DR4, DQ alleles (vary by population); GWAS signals at the immunoglobulin heavy-chain (*IGH*) and HLA-DQ loci.
- **Non-HLA immune genes**: cytokine polymorphisms (TNF-α, TGF-β1, IL-1/IL-10), mannose-binding lectin, Toll-like receptors, Fcγ receptors, and the historical **D8/17 B-cell alloantigen** (marker of susceptibility).

Proportion of ARF that progresses to RHD

- After a **first ARF episode: roughly 30–45%** develop detectable RHD; figures of **40–65%** are cited for "clinically recognisable" RHD over time.
- If **carditis is present at the first attack**, progression rises to **~60–70%**.
- **Recurrences** are the strongest amplifier — each recurrent episode adds incremental valve damage; this is why secondary prophylaxis is pivotal.

How long from ARF to RHD?

Typically **5–15 years** from the initial attack to established chronic RHD. **Mitral stenosis** develops latest (commissural fusion over 10–20+ yr in temperate regions). In highly endemic tropical settings an accelerated course ("juvenile mitral stenosis") can appear within a few years owing to repeated recurrences.

Pharyngitis aetiology — GAS, bacterial vs viral, and the bacterial differential

Question	Answer (approximate)
Proportion of pharyngitis that is GAS	**Children ~20–30%**; **adults ~5–15%** (~10%). GAS is the leading *treatable bacterial* cause.
Bacterial vs viral overall	**Majority viral** (~70–85% of all sore throats): rhinovirus, adenovirus, coronavirus, influenza/parainfluenza, EBV, enterovirus, HSV, and acute HIV.
Bacterial differential (beyond GAS)	Groups C & G β-haemolytic streptococci; *Fusobacterium necrophorum* (Lemierre's, adolescents); *Arcanobacterium haemolyticum* (scarlatiniform rash, teens); *Mycoplasma pneumoniae*; *Chlamydophila*; *Neisseria gonorrhoeae*; rarely *Corynebacterium diphtheriae*.

> Only GAS (and rarely groups C/G) carries ARF risk; treatment is targeted at GAS to interrupt the immunologic cascade and reduce transmission.

2. Pathogenesis & natural history of RHD

Pathogenesis — molecular mimicry and the "two-hit" autoimmune cascade

1. **Trigger**: GAS pharyngitis presents conserved epitopes — the surface **M protein** and the **group A carbohydrate (N-acetyl-β-D-glucosamine, GlcNAc)**.
2. **Cross-reactive immunity (molecular mimicry)**: in susceptible hosts, antibodies and CD4+ T cells raised against these epitopes cross-react with host proteins that share structural homology — **cardiac myosin, tropomyosin, laminin, vimentin, keratin** and the **valvular endothelium**.
3. **Valvular initiation**: anti-streptococcal/anti-myosin antibodies bind the valve surface endothelium → up-regulation of **VCAM-1** → recruitment and infiltration of activated **CD4+ T cells** through the endothelium into the avascular valve.
4. **Amplification**: local inflammation, neovascularisation and **epitope spreading** broaden the autoimmune attack; granulomatous **Aschoff bodies** (with Anitschkow "caterpillar" cells and Aschoff giant cells) form in the myocardium — the histological hallmark.
5. **Tissue injury**: a **pancarditis** (endo-, myo-, pericarditis) occurs, but it is the **endocardial/valvular** damage that persists — verrucae along the leaflet coaptation lines acutely, then with recurrences progressive **leaflet thickening, commissural fusion, chordal shortening/fusion, fibrosis and calcification**.

Natural history

- **Acute carditis**: regurgitation predominates (mitral > aortic) due to leaflet inflammation and annular dilation; may resolve, persist, or progress.
- **Latent/sub-clinical phase**: echo-detectable valve changes without a murmur — the window for screening and secondary prophylaxis.

- **Recurrent ARF**: each episode adds cumulative damage; mitral stenosis emerges late via commissural fusion.
- **Chronic RHD**: mixed mitral disease, aortic involvement, functional tricuspid regurgitation.
- **Complications**: atrial fibrillation, left atrial enlargement, pulmonary hypertension, heart failure, systemic thromboembolism/stroke, infective endocarditis, and premature death.

Valve involvement	Approximate frequency
Mitral valve (alone or combined)	Most common — involved in ~≥90% of RHD
Mitral + aortic	~20–30%
Aortic alone	Less common
Tricuspid	Usually *functional* and secondary to left-sided disease/PH; organic tricuspid rare
Pulmonary	Rare

3. Prevention at all levels (primordial → secondary) — Australian guideline

Level	Target	Interventions
Primordial	Underlying social determinants (before risk factors arise)	Reduce household crowding; improve housing, water/sanitation, hygiene and nutrition; health literacy; equitable healthcare access & economic development.
Primary	Prevent the *first* ARF	Prompt recognition and **adequate antibiotic treatment of GAS pharyngitis** (and skin infection in endemic settings), ideally within 9 days of symptom onset.
Secondary (most cost-effective)	Prevent *recurrent* ARF in those with prior ARF/RHD	Long-term continuous **benzathine penicillin G (BPG) every 4 weeks** (or oral penicillin V if injections declined); register/recall programs.
Tertiary	Limit morbidity of established RHD	Heart-failure & AF therapy, anticoagulation, valve intervention (PBMC/surgery), endocarditis prevention, pregnancy planning, specialist follow-up.

Key specifics from the RHDAustralia guideline (3rd ed.)

- **BPG dose**: 1,200,000 U IM (≥20 kg) or 600,000 U (<20 kg), every **28 days**; shorten to **21 days** only for documented breakthrough ARF despite adherence.
- **Duration of secondary prophylaxis** (by severity):

Category	Minimum duration
ARF, no carditis / no RHD	5 years after last ARF, or until age 21 (whichever is longer)
Mild–moderate RHD / healed carditis	Until age 35 (or 10 years, whichever is longer)
Severe RHD / after valve surgery	Until age 40 or longer — often lifelong

- **Register-based control programs** with active recall and priority reminders are central to delivery and adherence.
- Pain-reduction strategies for injections (warming, slow injection, lidocaine-reconstituted BPG where permitted) to support adherence.

4. Updates on vaccination — where are we, and prospects

No licensed GAS vaccine yet. Persistent obstacles: enormous antigenic diversity (>200 emm types), the historical safety concern of *molecular mimicry / autoimmunity*, absence of an accepted correlate of protection, and the need for efficacy trials in endemic settings.

Approach	Candidates / status (2025–2026)
M-protein based (type-specific)	**StreptAnova** (30-valent): Phase 1 completed — safe, immunogenic, opsonophagocytic antibodies, no autoimmunity signal.
Minimal M-protein epitope (conserved)	**J8 / p*17**-based peptide conjugates (e.g. p*17-K4S2, J8-K4S2 combined with a SpyCEP epitope) — early-phase clinical evaluation; designed to avoid cardiac cross-reactivity.
Non-M / conserved antigens	Targets include **SpyAD, SpyCEP, streptolysin O, C5a peptidase (ScpA), group A carbohydrate (GAC) conjugates, PrsA**; multicomponent combinations in preclinical/early development.
Next-generation platforms	**mRNA** vaccines encoding multiple conserved antigens (preclinical, 2025); recombinant glycoconjugate platforms.

Enabling infrastructure & prospects

- **SAVAC** (Strep A Vaccine Global Consortium) and **ASAVI** (Australian Strep A Vaccine Initiative); WHO *Preferred Product Characteristics* and a roadmap.
- **Controlled Human Infection Models (CHIM)** for GAS pharyngitis (established in Australia) can accelerate early efficacy read-outs.
- **Outlook**: momentum is real (post-pandemic GAS resurgence, invasive-GAS attention, mRNA platforms), but a licensed product remains several years away and is limited chiefly by *sustained funding* and trial capacity in endemic regions.

5. Why are females more susceptible? (new 2025 paper)

> **New evidence — Mutarelli et al., *Frontiers in Cardiovascular Medicine* 2025** (systematic review & updated meta-analysis, 58 studies, 215,552 children/adolescents screened by echo). Echo-detected RHD prevalence ~24/1000; **males had ~30% lower risk (RR 0.70)** for any RHD and ~29% lower (RR 0.71) for definite RHD. Crucially, the **female predominance appears *before* established valve lesions and in childhood**, arguing for an *intrinsic biological* susceptibility rather than purely pregnancy-related or detection effects.

ARF incidence is roughly **equal between sexes in children**, yet clinically manifest chronic RHD is **2:1 to 4:1 female**. Proposed (overlapping) mechanisms:

- **Immuno-biological / hormonal**: females mount stronger humoral and autoimmune responses; X-linked immune genes (e.g. TLR7) can escape X-inactivation; oestrogen modulates immune signalling. This fits the new finding that the sex gap exists early.
- **Pregnancy as a haemodynamic & immunologic "stress test"**: increased blood volume, heart rate and a relatively immunotolerant state can *unmask/accelerate* mitral stenosis and reveal latent disease in reproductive-age women.
- **General female preponderance of autoimmunity.**
- **Gendered access to care**: in some settings girls/women receive less prompt pharyngitis treatment and less reliable secondary prophylaxis — a modifiable inequity (compounds, but no longer the sole explanation given the 2025 data).

6. Time from ARF to RHD

Typically 5–15 years. Regurgitant lesions may be evident acutely or within a few years; **mitral stenosis is the latest manifestation** (commissural fusion over 10–20+ years in temperate climates). In high-burden tropical settings, repeated recurrences produce an accelerated "**juvenile mitral stenosis**" within a few years of the first attack.

7. Rapid tests for GAS pharyngitis

Test	Performance & use
Rapid antigen detection test (RADT)	Lateral-flow/EIA/optical immunoassay detecting group A carbohydrate. **Specificity high (~95%)**, **sensitivity variable (~70–90%**, newer assays ~85–90%). A positive RADT confirms; in *children* a negative result should be **backed up by throat culture** (IDSA), whereas backup is generally unnecessary in adults.
Molecular / NAAT (PCR, isothermal)	**Sensitivity & specificity ~95–99%**; can replace culture and give same-visit results. Caveat: cannot distinguish *carriage* from true infection.
Throat culture	Historic **gold standard** (24–48 h); still the confirmatory/backup test.

> Use a clinical score (Centor / McIsaac) to decide whom to test. In high-incidence/endemic populations (e.g. Australian guidance), routine RADT is *not* recommended — clinical suspicion plus empirical treatment is often preferred, because the consequence of missing GAS (ARF) is severe.

8. Treatment of GAS pharyngitis

Scenario	Regimen
First line (oral)	**Penicillin V** 500 mg (child 250 mg) PO 2–3×/day × **10 days**; **or amoxicillin** 50 mg/kg once daily (max 1000 mg) × 10 days.
Single-dose / adherence-limited	**Benzathine penicillin G** IM — 1.2 MU (≥27 kg) or 0.6 MU (<27 kg), once.
Penicillin allergy (non-anaphylactic)	First-generation cephalosporin (e.g. **cephalexin**) × 10 days.
Penicillin allergy (anaphylactic)	**Azithromycin** 12 mg/kg/day × 5 days, clarithromycin, or **clindamycin** — note rising macrolide resistance.

- A full **10-day course** (azithromycin 5 days) is required for eradication; treatment started **within 9 days** of onset still prevents ARF.
- **No penicillin resistance** has been documented in GAS — penicillin remains drug of choice.

9. ASO titres — duration of elevation & diagnostic threshold

- **Kinetics**: ASO rises within ~1 week, **peaks at 3–5 weeks**, then falls; commonly returns toward baseline by ~6 months but can stay elevated up to **6–12 months**.
- **Anti-DNase B** rises later (peak 6–8 weeks) and **stays elevated longer (many months)** — more useful for late presentations, chorea, and following skin infection. Using **both** ASO + anti-DNase B detects ~95% of recent GAS infections.
- **What counts as "elevated"**: the upper limit of normal is **age- and population-specific** (defined as the local ~80th centile). Common reference points: ASO **>200 IU/mL in adults** and **>320–333 IU/Todd units in school-age children**.
- **Diagnosing recent infection**: a single high titre indicates *recent* (not necessarily current) infection. The most reliable evidence is a **rising titre — a ≥2-fold (2-dilution) increase** between acute and convalescent sera drawn 2–4 weeks apart.

> Titres reflect prior infection and can be raised by carriage or unrelated recent GAS exposure; they are *not* used to diagnose acute pharyngitis or to monitor treatment response.

10. Management of RHD (overview)

1. **Confirm & stage**: echocardiography using WHF criteria; define lesion(s), severity, chamber sizes, PH.
2. **Secondary antibiotic prophylaxis** — the foundation (see Q3, Q12).
3. **Medical therapy**: diuretics for congestion; rate control + anticoagulation for AF; β-blockers to improve diastolic filling in MS; ACEi/ARB/MRA for regurgitant lesions with LV dysfunction (caution in pure MS); manage PH.
4. **Anticoagulation** for mechanical valves, AF, prior embolism, or LA thrombus — **VKA (warfarin)**; DOACs are *contraindicated* in rheumatic MS/mechanical valves (INVICTUS).
5. **Infective endocarditis prevention**: meticulous dental/skin hygiene; antibiotic prophylaxis for the highest-risk groups per current AHA guidance (prosthetic valve, prior IE).
6. **Intervention**: PBMC for suitable mitral stenosis; surgical repair/replacement when indicated (see Q16, Q20, Q21).
7. **Holistic care**: pregnancy/contraception counselling, vaccination (influenza, pneumococcal), comorbidity control, and multidisciplinary register-based follow-up.

11. Management of acute rheumatic fever

- **Confirm diagnosis** (revised Jones), admit, rest during active carditis.
- **Eradicate GAS**: penicillin (single IM BPG or 10-day oral) for *all* patients regardless of throat-swab result — this also **starts secondary prophylaxis**.
- **Arthritis / fever**: high-dose **aspirin** (50–100 mg/kg/day) or **naproxen**. *Withhold anti-inflammatories until the diagnosis is secure* in undifferentiated arthralgia, so the migratory polyarthritis is not masked. Anti-inflammatories relieve symptoms but **do not change cardiac outcome**.
- **Carditis / heart failure**: treat HF (diuretics, ACEi as tolerated); **corticosteroids** (prednisone) are commonly used for moderate–severe carditis (evidence weak); valve surgery rarely needed for acute severe regurgitation.
- **Sydenham chorea**: usually self-limiting; for disabling cases use **carbamazepine or valproate** (avoid valproate in pregnancy), occasionally corticosteroids.
- **Then**: echocardiography in all cases; commence/secure long-term secondary prophylaxis; arrange follow-up and registration.

12. Evidence: oral Penicillin V vs IM benzathine penicillin

- **Historical & Cochrane evidence** favours **IM BPG over oral penicillin** for secondary prophylaxis — fewer GAS infections and ARF recurrences, largely because IM gives sustained drug levels and removes daily-adherence dependence.
- **Standard of care** (AHA, WHF, RHDAustralia): **4-weekly BPG** (3-weekly in high-risk/breakthrough). Oral penicillin V is reserved for those who decline injections.
- **GOAL trial** (NEJM 2022, Uganda): in children with *latent* RHD, 4-weekly IM BPG **reduced 2-year progression to ~0.8% vs 8.4%** with no prophylaxis — the first RCT proving benefit of prophylaxis in screen-detected disease.
- **GOALIE trial** (ongoing, NCT05693545): a **non-inferiority RCT of oral penicillin V vs IM BPG** for latent RHD. If oral proves non-inferior it could shift practice toward an acceptable, needle-free option — results awaited.

13. Ongoing research in RHD

- **GOALIE** — oral vs IM penicillin for latent RHD (above).
- **Reformulated long-acting BPG** — high-dose subcutaneous infusion / extended-interval formulations to reduce injection frequency and pain (PK-driven programs).
- **Echo screening at scale** — handheld and **AI-assisted echocardiography**, task-shifting to non-specialists, and the open question of whether population screening improves outcomes.

- **GAS vaccine development** and **CHIM** efficacy models (Q4).
- **Penicillin-allergy de-labelling** to widen access to first-line prophylaxis.
- **Pathogenesis, genetics & biomarkers** — GWAS (IGH, HLA-DQ), point-of-care diagnostics, and the biology of sex predisposition (Q5).
- **Therapeutics evidence** — INVICTUS (VKA superior to rivaroxaban in rheumatic AF) now informs anticoagulation; valve-repair techniques and prosthesis choice in young LMIC populations under study.
- **Implementation & policy** — registers, the 2018 WHA resolution on RHD, and health-system strengthening.

14. Theories for haemoptysis in mitral stenosis

1. **"Pulmonary apoplexy"** (sudden, profuse) — rupture of dilated, thin-walled **bronchial/bronchopulmonary venous anastomoses** under high left-atrial/pulmonary-venous pressure. Classically occurs *earlier* in disease (high LA pressure, still-low PVR) and may paradoxically lessen as pulmonary vascular resistance rises and "protects" the bed.
2. **Pulmonary oedema** — pink, frothy, blood-tinged sputum from alveolar–capillary disruption.
3. **Pulmonary infarction** — from pulmonary embolism (AF, low output, immobility); a later complication.
4. **Chronic bronchitis / bronchial mucosal congestion** ("winter bronchitis") — blood-streaked sputum.
5. **Lower respiratory infection** superimposed on congested lungs.

15. Wilkins score & other PBMC scores

Wilkins (Abascal) echocardiographic score

Four morphological features, each graded **1–4 → total 4–16**:

Component	Graded 1 (best) → 4 (worst)
Leaflet mobility	Highly mobile (only tips restricted) → minimal/no diastolic movement
Leaflet thickening	Near-normal (4–5 mm) → marked thickening (>8–10 mm)
Subvalvular thickening	Minimal, just below leaflets → extensive thickening/shortening to papillary muscles
Calcification	Single area → extensive calcification throughout leaflets

> **Interpretation**: total **≤8 → favourable** for PBMC (better immediate results, lower restenosis); **>8 less favourable**; **≥10–12 poor** outcome.

Limitations & alternative scores

- **Wilkins limitation**: weights features equally and does not capture *commissural calcium distribution* or pre-existing MR — both strong predictors of poor outcome.
- **Cormier score** — three anatomic groups based on mobility, valve calcification and subvalvular disease.
- **Nunes / "cardiac" score** — integrates valve area, maximal leaflet displacement, commissural-area ratio and subvalvular thickening; better predicts suboptimal results.
- **Commissural calcification scoring** (e.g. Sutaria) — asymmetric/commissural calcium predicts failure and procedural MR.

16. Indications & contraindications for PBMC

Indications (ACC/AHA & ESC)

- **Class I**: *symptomatic* severe MS (MVA ≤1.5 cm²) with **favourable valve morphology**, no LA thrombus, and no/mild MR (≤2+).
- **Class IIa/IIb**: *asymptomatic* severe MS with favourable anatomy plus pulmonary hypertension (PASP >50 mmHg), **new-onset AF**, or planned **pregnancy / major non-cardiac surgery**.
- Symptomatic patients with less favourable anatomy when surgical risk is high.

Contraindications

Absolute	Relative / unfavourable
Left atrial / LAA **thrombus**; moderate–severe MR (>2+ / grade III–IV); MVA >1.5 cm² (not severe)	Severe or bicommissural calcification; absence of commissural fusion; high Wilkins score (>8–10); concomitant severe aortic/tricuspid disease or CAD needing surgery (favour surgery)

17. Revised Jones criteria (2015) & frequency of manifestations

Diagnosis (with evidence of antecedent GAS infection): **initial ARF** = 2 major *or* 1 major + 2 minor; **recurrent ARF** = 2 major, *or* 1 major + 2 minor, *or* 3 minor. The 2015 revision is **risk-stratified** (low- vs moderate/high-risk populations) and admits

subclinical (echo) carditis.

	Low-risk population	Moderate/high-risk population
Major	Carditis (clinical/subclinical), **polyarthritis**, chorea, erythema marginatum, subcutaneous nodules	As left, but arthritis also includes **monoarthritis or polyarthralgia**
Minor	Polyarthralgia; fever ≥38.5°C; ESR ≥60 mm/h and/or CRP ≥3.0 mg/dL; prolonged PR interval	Monoarthralgia; fever ≥38°C; ESR ≥30 mm/h and/or CRP ≥3.0 mg/dL; prolonged PR

Evidence of antecedent GAS: elevated/rising ASO or anti-DNase B, positive throat culture/RADT, or recent scarlet fever.

Major manifestation	Approximate frequency at presentation
Arthritis (migratory polyarthritis)	~60–75% — most common
Carditis	~50–70% (higher with routine echo / subclinical detection)
Sydenham chorea	~10–30% (often late; may occur in isolation)
Subcutaneous nodules	~0–10% (uncommon; associated with severe carditis)
Erythema marginatum	<6% (rare; with carditis)

18. Pathophysiology of pulmonary hypertension in RHD

Driven chiefly by **mitral stenosis** (and any left-sided lesion that raises LA pressure):

1. **Passive (post-capillary) component**: raised LA pressure is transmitted backward → pulmonary venous congestion.
2. **Reactive vasoconstriction**: pulmonary arteriolar constriction (initially reversible) limits capillary pressure but raises PVR.
3. **Fixed (pre-capillary) remodelling**: chronic elevation drives **medial hypertrophy and intimal fibrosis** of arterioles — a "second stenosis" — producing a fixed PVR rise.
4. **Right-heart consequences**: RV pressure overload → RV hypertrophy & dilatation → **functional tricuspid regurgitation** → right heart failure.

> As fixed PVR rises it partly "protects" the pulmonary capillaries (less acute oedema/haemoptysis) but at the cost of falling cardiac output and progressive RV failure.

19. Pathophysiology of atrial fibrillation in RHD

1. **Atrial stretch & pressure/volume overload** (MS & MR) → progressive **left atrial dilatation**.
2. **Structural remodelling**: atrial fibrosis (also from rheumatic inflammation involving the atria) creates conduction heterogeneity and slowed conduction.
3. **Electrical remodelling & triggers**: ectopy (incl. pulmonary-vein foci) on a substrate of stretch and fibrosis sustains **re-entry**.
4. **Haemodynamic consequences**: loss of atrial kick (worsens the MS gradient and output) and a rapid ventricular rate (shortens diastole, impairing LV filling in MS).
5. **Thromboembolism**: stasis in the dilated LA/LAA → thrombus → systemic embolism/stroke. Rheumatic AF carries high embolic risk — **VKA anticoagulation is indicated regardless of CHA_2DS_2-VASc score**.

20. Grading valve severity & thresholds for intervention

Lesion	"Severe" thresholds (integrated echo)	Key intervention triggers
Mitral stenosis	MVA ≤1.5 cm^2 (very severe ≤1.0); supportive mean gradient >5–10 mmHg; PASP elevated	Symptomatic severe MS → PBMC (favourable anatomy) or surgery; asymptomatic severe + PH/new AF/pregnancy (IIa/IIb)
Mitral regurgitation	EROA ≥0.40 cm^2; regurgitant volume ≥60 mL; RF ≥50%; vena contracta ≥0.7 cm	Symptomatic severe primary MR; or LVEF ≤60% / LVESD ≥40 mm → surgery (**repair preferred**)
Aortic stenosis	AVA ≤1.0 cm^2; V_{max} ≥4 m/s; mean gradient ≥40 mmHg	Symptomatic severe AS; or LVEF <50% → valve replacement
Aortic regurgitation	Vena contracta >0.6 cm; EROA ≥0.30 cm^2; RVol ≥60 mL; holodiastolic aortic flow reversal	Symptomatic severe AR; or LVEF ≤55% / dilating LVESD → surgery

Always integrate symptoms, multiple echo parameters, chamber size/function, PH, and (where relevant) exercise testing rather than relying on a single number.

21. Choice of valve: mechanical vs bioprosthetic

	Mechanical	Bioprosthetic
Durability	Excellent — usually lifelong	Limited — structural deterioration; faster in the young → reoperation
Anticoagulation	**Lifelong VKA required**	No long-term VKA (short initial period ± antiplatelet)
Favoured when	Younger (<~50–60 yr); already needing/able to take anticoagulation; low bleeding risk; high reoperation risk	Older (>~65–70 yr); bleeding risk or unable/unwilling to anticoagulate; **women planning pregnancy**; limited life expectancy; patient preference
Main downside	Bleeding & thrombosis risk; warfarin burden & monitoring	Reoperation for degeneration

> In young women in RHD-endemic regions, the **pregnancy question is pivotal**: a mechanical valve commits to warfarin, which is hazardous in pregnancy (Q24). Use shared decision-making and a valve/heart team.

22. Anticoagulation in VHD — target INR & time in therapeutic range

Setting	Agent & target INR
Mechanical aortic (newer-gen, no risk factors)	VKA, **INR 2.5** (2.0–3.0)
Mechanical aortic + risk factors / older valve	VKA, **INR 3.0** (± low-dose aspirin)
Mechanical mitral	VKA, **INR 3.0** (2.5–3.5)
Rheumatic MS + AF / prior embolism / LA thrombus	**VKA, INR 2.0–3.0 — DOACs contraindicated** (INVICTUS: VKA superior to rivaroxaban)
Bioprosthetic	Anticoagulation ~3–6 months (esp. mitral), then aspirin; DOAC/none acceptable if no other indication

> **Time in therapeutic range (TTR, Rosendaal method)**: aim **>65–70%**. Higher TTR means fewer thromboembolic *and* bleeding events; poor TTR (common in resource-limited settings) is a major driver of complications — quality of control matters as much as the target itself.

23. Warfarin — drug–drug & food–drug interactions

Effect	Examples
Potentiate (↑INR / bleeding)	Amiodarone, metronidazole, azole antifungals (fluconazole), TMP-SMX, ciprofloxacin/macrolides, many antibiotics, **NSAIDs & aspirin** (also direct bleeding), SSRIs, some statins, allopurinol, omeprazole, acute alcohol binge, acute febrile illness, hyperthyroidism, cranberry juice.
Inhibit (↓INR)	Rifampicin, carbamazepine, phenytoin, barbiturates, **St John's wort**, high vitamin-K intake, chronic alcohol use, hypothyroidism.
Food / diet	**Vitamin-K-rich foods** (kale, spinach, broccoli, green leafy vegetables) — keep intake *consistent*, do not avoid; limit alcohol; caution with grapefruit, cranberry, green tea.

CYP2C9 and VKORC1 polymorphisms alter dose requirements. Counsel on dietary consistency, regular INR monitoring, and reviewing every new medicine (including OTC and herbal).

24. Anticoagulation in pregnancy (mechanical valves & RHD)

Mechanical valves carry the highest maternal thrombotic risk; every option trades maternal valve-thrombosis risk against fetal harm.

Agent	Profile
Warfarin (VKA)	**Most effective** against valve thrombosis but crosses placenta → **embryopathy (esp. weeks 6–12)**, fetal loss, fetal bleeding — risk is dose-related (notably >5 mg/day).
LMWH	Fetal-safer; needs **anti-Xa monitoring** (peak ~0.8–1.2 U/mL); higher valve-thrombosis risk if under-dosed.
UFH	Does not cross placenta; used peripartum.

Typical ESC-aligned strategy

- **Weeks 6–12**: if warfarin ≤5 mg/day, may continue with counselling (lower embryopathy risk) *or* switch to monitored LMWH.
- **2nd–early 3rd trimester**: warfarin (or dose-adjusted LMWH) for best valve protection.
- **~36 weeks / pre-delivery**: switch to LMWH then **IV UFH**; stop before delivery to avoid fetal/neonatal bleeding; resume

post-partum.

- Low-dose aspirin may be added; manage rheumatic MS with β-blockers/diuretics and PBMC if severely symptomatic.

> Manage in a **multidisciplinary pregnancy heart team**; classify risk with the modified WHO system (Q25).

25. Modified WHO (mWHO) maternal cardiovascular risk classification

Class	Risk	Examples relevant to RHD
I	No detectable increase in maternal risk	Small/mild, repaired lesions
II	Small increase	Mild mitral/aortic disease, unoperated minor lesions
II–III	Intermediate	Moderate left-heart obstruction; mild LV impairment
III	Significantly increased — expert care, intensive monitoring	**Mechanical valve**; moderate mitral stenosis; systemic RV
IV	**Pregnancy contraindicated**; if it occurs, discuss termination	**Severe (symptomatic) mitral stenosis**; severe pulmonary hypertension/Eisenmenger; severe systemic-ventricular dysfunction (EF<30% / NYHA III–IV); severe symptomatic AS

Estimated maternal cardiac event rates rise across classes (roughly <5% in I to >40–100% in IV). The system guides counselling, the decision to proceed, and the intensity/location of antenatal care.

26. Choice of contraception in RHD/VHD

- **Avoid combined (oestrogen-containing) methods** in high-risk patients — mechanical valve, AF, prior thromboembolism, pulmonary hypertension, cyanosis — due to thrombotic risk.
- **Preferred**: progestogen-only **long-acting reversible contraception (LARC)**:
  - **Levonorgestrel IUD** (Mirena) — highly effective, low thrombotic risk, and reduces menstrual bleeding (useful for women on anticoagulation).
  - **Etonogestrel implant** — highly effective, oestrogen-free.
  - Progestogen-only pill / DMPA injection — acceptable; note bone/weight considerations with DMPA.
- **Copper IUD** — non-hormonal and effective, but can increase menstrual bleeding (caution on anticoagulants). Perform insertion where vasovagal events can be managed (relevant in severe PH).
- **Permanent options** (tubal ligation / vasectomy of partner) when family is complete or pregnancy is contraindicated — weigh peri-procedural risk in severe disease.
- Use **UK/WHO Medical Eligibility Criteria (MEC)** categories; consider dual methods and access to emergency contraception.

27. Management of heart failure in RHD

1. **Define the mechanism**: which valve lesion(s), LV vs RV failure, presence of AF and PH.
2. **General measures**: salt/fluid moderation; **treat precipitants** — anaemia, infection, new AF, pregnancy, infective endocarditis, and non-adherence to prophylaxis.
3. **Congestion**: **loop diuretics** are the mainstay (lower LA pressure in MS; relieve congestion in regurgitant lesions).
4. **Mitral stenosis specifics**: **rate control** (β-blocker, ivabradine, non-DHP CCB, or digoxin in AF) to lengthen diastole; anticoagulate if AF; **relief of obstruction (PBMC/surgery) is the definitive therapy**. Avoid aggressive vasodilation.
5. **Regurgitant lesions with LV dysfunction**: guideline-directed therapy — **ACEi/ARB, β-blockers, MRA** (afterload reduction helps MR/AR; use cautiously/avoid in pure MS).
6. **AF**: rate or rhythm control *plus* VKA anticoagulation.
7. **Decompensation**: IV diuretics, treat pulmonary oedema, address PH/RV failure.
8. **Definitive correction**: timely valve intervention *before* irreversible LV dysfunction or fixed PH develops.
9. **Ongoing**: continue secondary prophylaxis, IE prevention, vaccination, comorbidity control, and pregnancy/contraception planning; advanced therapies (devices/transplant) only in end-stage disease.

Key references (selected, 2022–2026)

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Prepared as a study aid. Always cross-check doses and thresholds against the current local protocol and the most recent guideline editions before clinical use. Several frequency figures (e.g. Jones manifestation rates, progression proportions) are approximate ranges that vary by population and study methodology.